

ELECTRONICS AND INSTRUMENTATION ENGINEERING

Date	1 st April, 2025	Maximum Marks	10+50
Course Code	EI243AI	Duration	120 Mfin
Sem	IV	CIE (Quiz & Test) -1	
MICROCONTROLLER AND PROGRAMMING			

Sl. No.	PART-A	M	BT	CO
1.1	Unit in a processor is responsible for arithmetic and logical operations.	1	1	1
1.2	Architecture has separate memory spaces for instructions and data.	1	1	1
1.3	Which register in ARM processors holds the application program status information?	1	1	1
1.4	Represent following number in little-endian and big-endian format 0x12345678	2	1	2
1.5	What is the operation of assembly instruction STR r0, [r1, #12]	1	1	1
1.6	What is the purpose of the Program Counter (PC) and stack pointer (SP) in processor execution?	1	1	1
1.7	PRE R0 = 0x00000000 R1 = 0x00009000 Mem32[0x00009000] = 0x01010101 Mem32[0x00009004] = 0x02020202 LDR R0, [R1, #4] What is the value of R0 after execution of above instruction	2	2	2
1.8	What is the ARM instruction set that combines 16-bit and 32-bit instructions?	1	2	1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

B1 - Blooms Taxonomy, CO-Course Outcomes, and Learning												
Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz 1	Max Marks (10)	6	4	-	-	6	4	-	-	-	-



RV College of
Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India
+91-080-68188100 | www.rvce.edu.in

Go, change the world®

Academic year 2024-2025 (EVEN Sem)

Sl. No.	PART- B	M	BT	CO
1.a	Explain the role of the control unit in a processor. Discuss how the control unit manages the execution of instructions and coordinates the operations of the ALU, registers, and memory.	6	1	1
1.b	Describe the main difference between CISC and RISC instruction set architectures.	4	1	2
2.a	Discuss the role of the APSR (Application Program Status Register) in the ARM Cortex M architecture and its significance in managing program states.	6	1	2
2.b	Describe the Thumb-2 instruction set of the ARM Cortex M architecture. How does it enhance performance and reduce memory usage compared to standard ARM instruction sets?	4	3	3
3.a	Briefly explain the difference between the ARM Cortex A, R, and M families of processors.	6	2	2
3.b	Represent 125.625 floating number in standard single precision representation.	4	2	3
4.a	Analyze the ARM Cortex M architecture and describe how its efficient instruction set and low-power features make it suitable for real-time and embedded applications.	4	3	2
4.b	List and explain the various operation modes and states in the ARM Cortex M architecture.	6	2	1
5.a	Write an assembly level program to exchange of block of RAM memory.	6	3	3
5.b	Explain usage LDR and STR assembly instruction with examples.	4	2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test 1	Max Marks (50)	12	20	18	-	16	20	14	-	-	-
